Veri Setinin Yüklenmesi

```
df = pd.read_csv("churn.csv")
df.head()
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  ...  DeviceProtection  TechSupport  StreamingTV  StreamingMovies        Contract PaperlessBilling  \
0  7590-VHVEG  Female              0     Yes         No       1           No  ...                No           No           No              No  Month-to-month              Yes
1  5575-GNVDE    Male              0      No         No      34          Yes  ...                Yes           No           No              No        One year               No
2  3668-QPYBK    Male              0      No         No       2          Yes  ...                 No           No           No              No  Month-to-month              Yes
3  7795-CFOCW    Male              0      No         No      45           No  ...                Yes          Yes           No              No        One year               No
4  9237-HQITU  Female              0      No         No       2          Yes  ...                 No           No           No              No  Month-to-month              Yes

   PaymentMethod  MonthlyCharges  TotalCharges Churn
0   Electronic check           29 ...
```

Exploratory Data Analysis (EDA)

In this section, we examine the overall structure of the dataset and conduct an initial exploratory data analysis (EDA). Using basic pandas functions, we review the dataset's dimensions, data types, and summary statistics to identify potential patterns, anomalies, and data quality issues.

```
df.shape
```

```
(7043, 21)
```

```
df.info()
```

```
<class pandas.DataFrame>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21):
 #   Column            Non-Nulls Dtype
---  ------            --------- -----
 0   customerID        7043 ...
```

```
df.describe()
```

```
       SeniorCitizen       tenure  MonthlyCharges
count    7043.000000  7043.000000     7043.000000
mean        0.162147    32.371149       64.761692
std         0.368612    24.559481       30.090047
min         0.000000     0.000000       18.250000
25%         0.000000     9.000000       35.500000
50%         0.000000    29.000000       70.350000
75%         0.000000    55.000000       89.850000
max         1.000000    72.000000      118.750000
```

```
df.nunique()
```

```
customerID        7043
gender            ...
```

```
df.isnull.sum()
```

```
customerID        0
gender            0
SeniorCitizen     0
Partner           0
Dependents        0
tenure            0
PhoneService      0
MultipleLines     0
InternetService   0
OnlineSecurity    0
OnlineBackup      0
DeviceProtection  0
TechSupport       0
StreamingTV       0
StreamingMovies   0
Contract          0
PaperlessBilling  0
PaymentMethod     0
MonthlyCharges    0
TotalCharges      0
Churn             0
dtype: int64
```

Veri Ön İşlemesi

```
df.drop("customerID", axis=1)
```

```
   gender  SeniorCitizen Partner Dependents  tenure PhoneService MultipleLines  ...  DeviceProtection  TechSupport  StreamingTV  StreamingMovies        Contract PaperlessBilling  \
0  Female              0     Yes         No       1           No            ...
```

```
df.TotalCharges = pd.to_numeric(df.TotalCharges, errors="coerce")
df.isnull.sum()
```

```
gender            0
SeniorCitizen     0
Partner           0
Dependents        0
tenure            0
PhoneService      0
MultipleLines     0
InternetService   0
OnlineSecurity    0
OnlineBackup      0
DeviceProtection  0
TechSupport       0
StreamingTV       0
StreamingMovies   0
Contract          0
PaperlessBilling  0
PaymentMethod     0
MonthlyCharges    0
TotalCharges      0
Churn             0
dtype: int64
```

```
df.replace({"Churn":{"No":0,"Yes":1}})
```

```
   gender  SeniorCitizen Partner Dependents  tenure PhoneService MultipleLines  ...  DeviceProtection  TechSupport  StreamingTV  StreamingMovies        Contract PaperlessBilling  \
0  Female              0     Yes         No       1           No            ...
```

Kategorik Sütunlar Üzerinde Dummy Dönüşümü

```
df.dtypes
```

```
gender                 object
SeniorCitizen           int64
Partner                object
Dependents             object
tenure                  int64
PhoneService           object
MultipleLines          object
InternetService        object
OnlineSecurity         object
OnlineBackup           object
DeviceProtection       object
TechSupport            object
StreamingTV            object
StreamingMovies        object
Contract               object
PaperlessBilling       object
PaymentMethod          object
MonthlyCharges         float64
TotalCharges           float64
Churn                  object
dtype: object
```

```
df.select_dtypes(object).columns
```

```
Index(['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines',
       'InternetService', 'OnlineSecurity', 'OnlineBackup',
       'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies',
       'Contract', 'PaperlessBilling', 'PaymentMethod', 'Churn'],
      dtype=object)
```

```
df = pd.get_dummies(df, drop_first=True)
df.head()
```

```
   SeniorCitizen  tenure  MonthlyCharges  TotalCharges  customerID_0002-ORFBO  customerID_0003-MKNFE  customerID_0004-TLHLJ  customerID_0011-IGKFF  customerID_0013-EXCHZ  customerID_0013-MHZWF  ...  PaymentMethod_Mailed check  Churn_Yes
0              0       1           29 ...
```

Lojistik Regresyon Modeli

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
model.score(X_test, y_test)
```

```
0.8006
```

Rastgele Kararlar Ormanı Sınıflandırıcısı Beklentisiz

```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(n_estimators=100, random_state=42)
```

```
model.score(X_test, y_test)
```

```
0.7892
```

Çapraz Doğrulama

```
from sklearn.model_selection import cross_val_score
cross_val_score(model, X, y, cv=5).mean
```

```
0.7888
```

Çapraz doğrulama, model performansını daha güvenilir bir şekilde ölçmek için veri setini birden fazla parçaya bölerek eğitim ve test aşamalarını farklı kombinasyonlarda gerçekleştirir. Bu yöntem, tek bir train-test ayrımına bağımlılığı azaltarak genelleme yeteneğinin daha objektif bir biçimde ölçülmesini sağlar.